

Decision-making guide

Capital Budgeting Microsimulation

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1. Simulation platform introduction

1.1 Game Arena

The gaming arena interface comprises various decision tabs that includes market, Project portfolio, Decide, Decision Checklist, Report, and Synopsis. Some sections must be completed first as they have an impact on other areas.

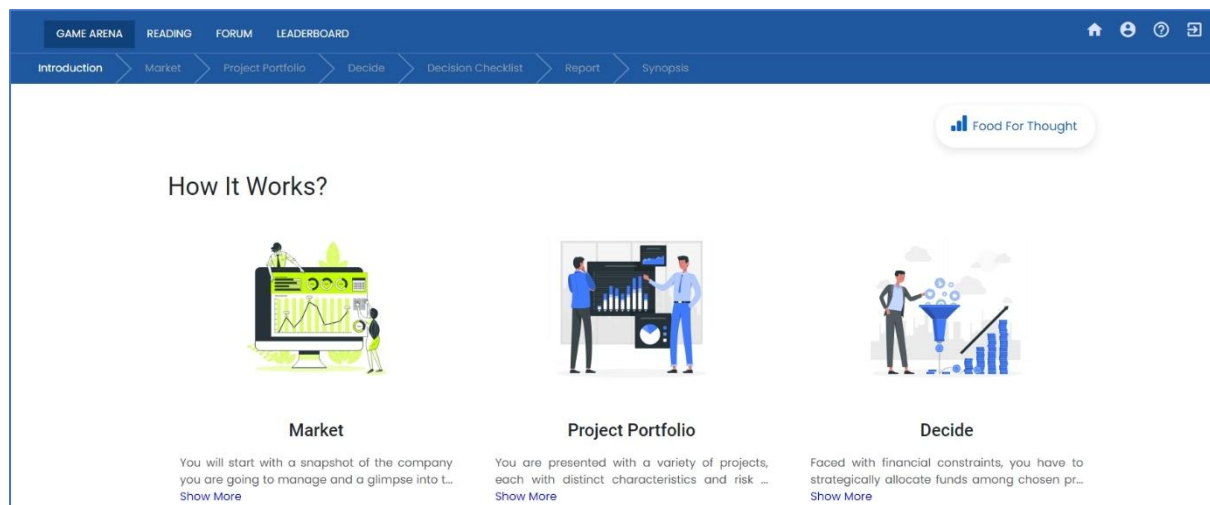


Figure 1 The Game Arena

1.2 Readings

This section provides essential materials necessary for participants to actively engage in the game. The main reading material consists of a guide for decision-making, supplemented by optional additional resources chosen by instructors. Moreover, a walkthrough video is accessible, offering in-depth information about the scenario featured in the simulation. This video presents insights into the industry, ongoing trends, obstacles, and the diverse array of decisions required.

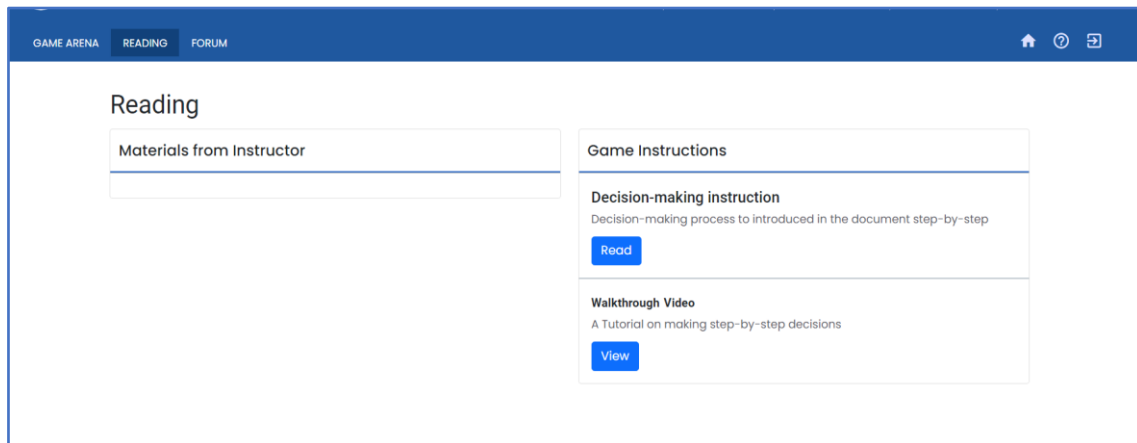


Figure 2 The Reading Tab

1.3 Forum

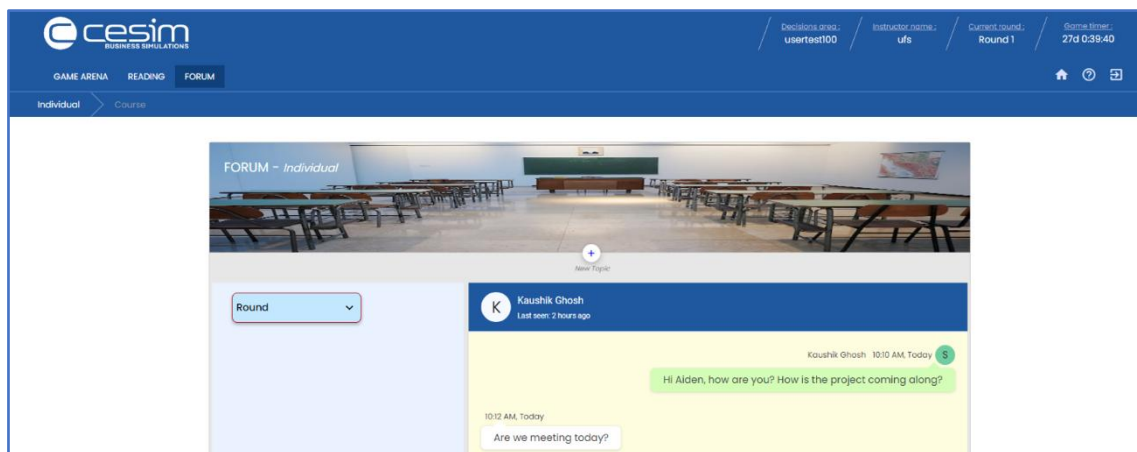


Figure 3 The Forum Tab

- The forums provide an excellent platform for participants to engage with their instructors or peers.
- There are two separate forums: the Individual Forum and the Course Forum. The Individual Forum allows participants to communicate directly with the instructor, while the Course Forum is accessible to all enrolled players.
- Instructors can view and respond to posts in both sections. Consequently, the Course Forum is an appropriate space for asking questions that can benefit the entire course, while the Individual Forum is the preferred channel for the individual to have a direct discussion with the professor.
- Participants will receive notifications for messages.

1.4 Main Objective & Winning Criteria

A New – Delhi based insurance software company operates across India, the Middle East, and Europe, offering innovative solutions to enhance efficiency and customer experience in the insurance sector. With 1,000 professionals, the company is managing 15 strategic projects focused on market expansion and operational improvements. Participants must strategically allocate the budget available by selecting the best projects that maximize financial performance while balancing risk and return.

Success is based on three Key Performance Indicators (KPIs):

1. Net Present Value (NPV) of Cash Stream – Higher is better.
2. Present Value (PV) of Cash Stream – Maximize returns.
3. Average Discount Rate Across Projects (%) – Maintain an optimal risk-return balance.

1.5 Successful decision-making flow

The sequence of decision-making unfolds as follows:

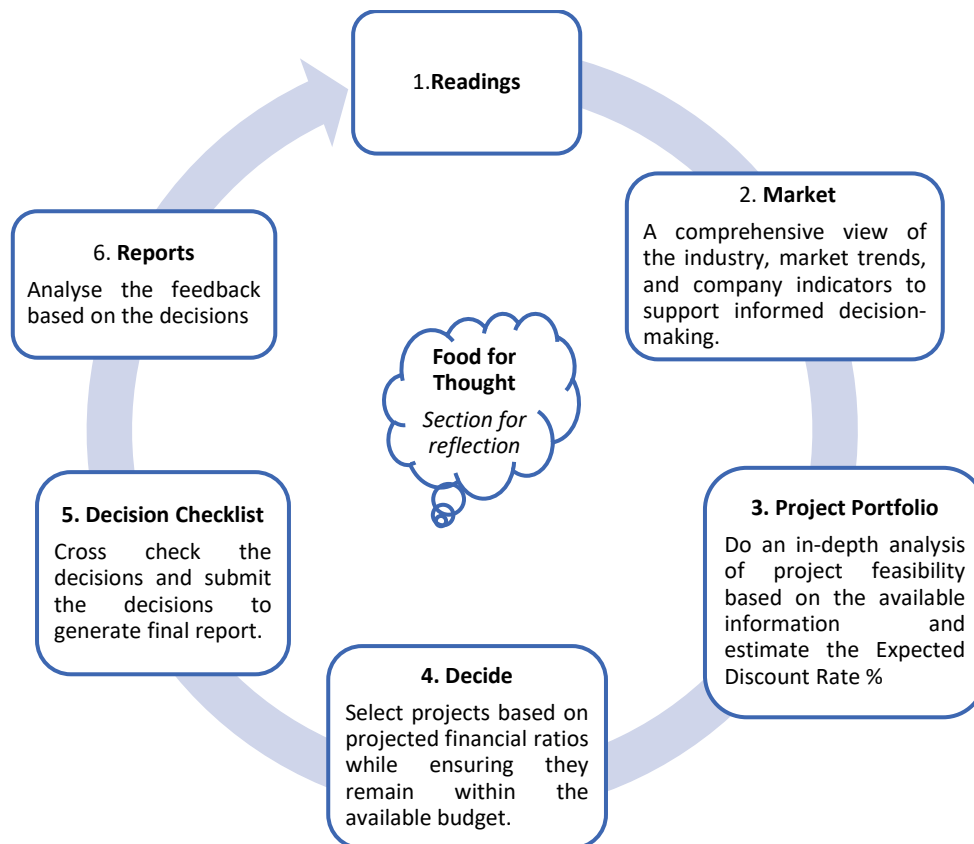


Figure 4 The Sequence of Decision-Making Process

2. Game Arena

2.1 Introduction

This page encompasses information detailing the functioning of the simulation and provides insights into each tab's decision-making.

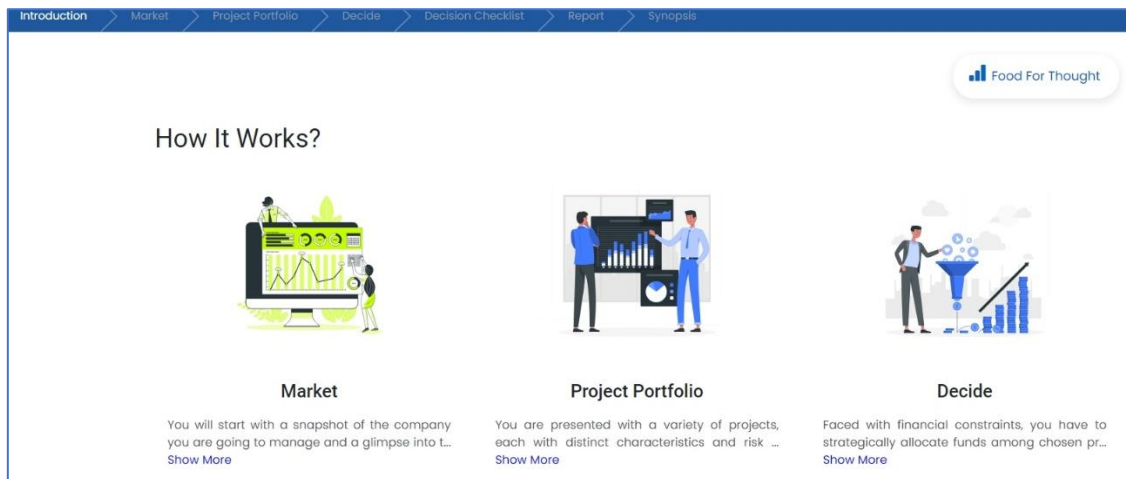


Figure 5 The Introduction Tab in the Game Arena

2.2 Food for Thought

This segment provides crucial aspects to consider, offering valuable insights for decision-making.

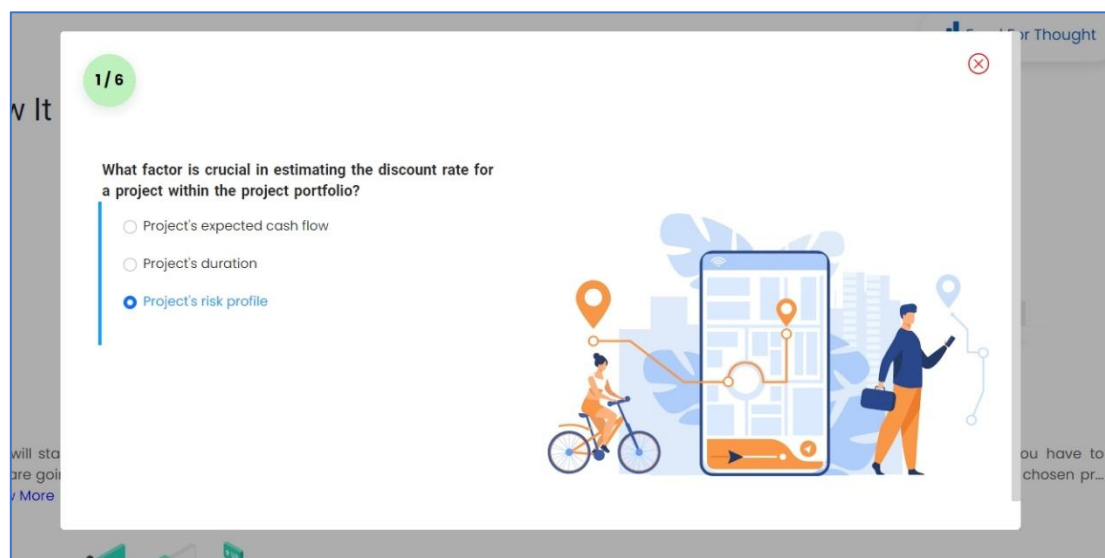


Figure 6 The Food for Thought Section in the Game Arena

2.3 Market

Prior to commencing the decision-making process, it is essential to carefully examine the "Memo" section. This section offers vital information on case scenario, and key parameters to consider when making decisions in each tab.

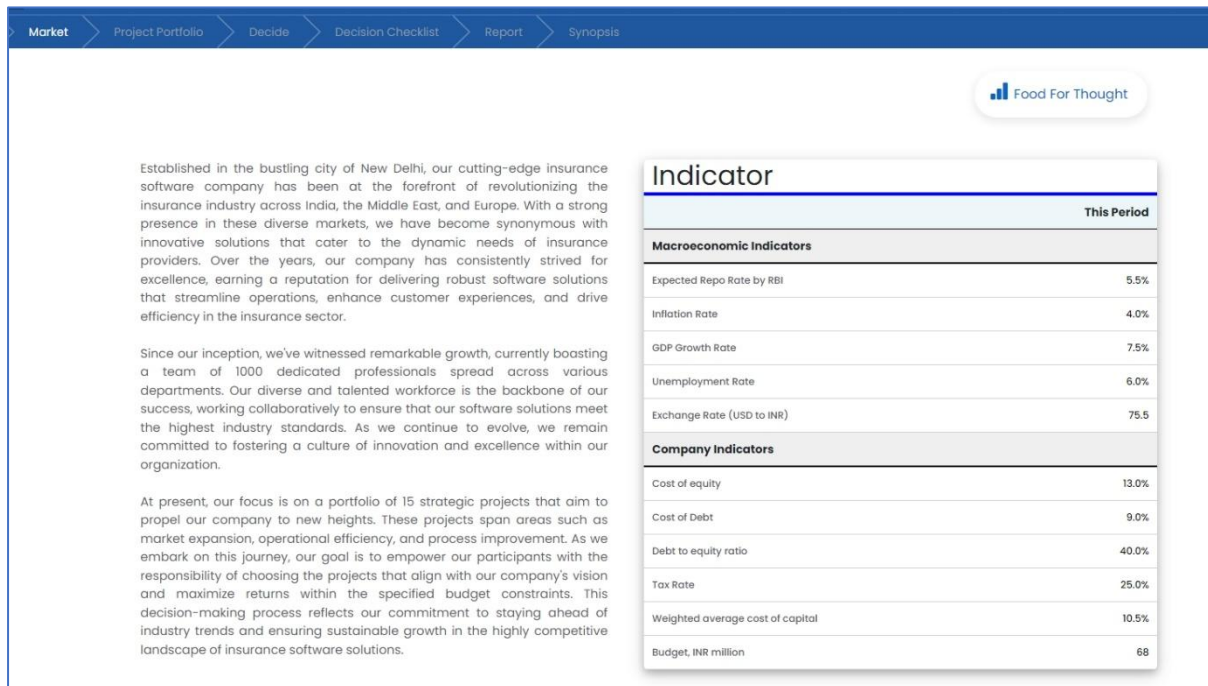


Figure 7 The Market Tab in the Game Arena

2.4 Project Portfolio

Under the Project Portfolio tab, participants have access to detailed information on various projects available for evaluation. Each project includes key details such as its objective, risk level, initial investment, expected cash flow per period, and project duration.

The projects span four key areas:

1. Market Expansion
2. Operational Efficiency
3. Process Improvement
4. Product Expansion

Based on this information, participants must determine the appropriate **Expected Discount Rate (%)** for each project, considering factors such as risk level, cost of capital, and potential returns.

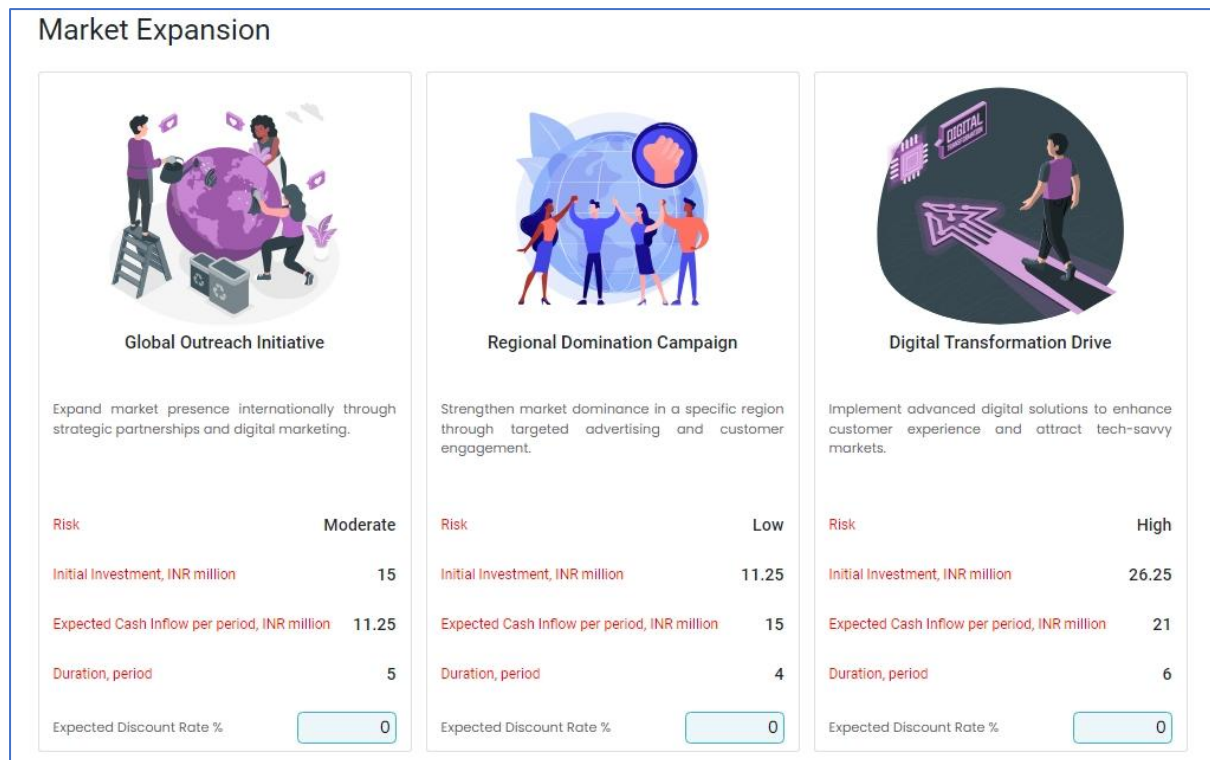


Figure 8 The Decision area Under the Project Portfolio Tab

2.5 Decide

Under the **Decide** tab, participants are presented with a table summarizing the various projects and option to choose from, along with key financial metrics such as:

- **Payback Period** – The time required to recover the initial investment from project cash flows.
- **Net Present Value (NPV)** – The present value of future cash flows minus the initial investment, indicating project profitability.
- **Accounting Rate of Return (ARR)** – The average annual accounting profit as a percentage of the initial investment.
- **Internal Rate of Return (IRR)** – The discount rate at which the project's NPV becomes zero, representing its expected return.
- **Profitability Index (PI)** – The ratio of the present value of future cash flows to the initial investment, measuring value creation per unit of investment.
- **Equivalent Annual Cost (EAC)** – The annualized cost of a project, helping compare projects with different lifespans.

Project Selection						
Decide	Area	Project	Investment, INR million	Discount rate %	Payback Period	ARR %
☐	Market Expansion	Global Outreach Initiative	15.00	7.00	1.33	375
☐	Market Expansion	Regional Domination Campaign	11.25	7.50	0.75	533
☐	Market Expansion	Digital Transformation Drive	26.25	7.50	1.25	480
☐	Market Expansion	Emerging Markets Expansion	30.00	7.50	1.33	525
☐	Operational Efficiency	Efficiency Overhaul Program	13.50	7.50	1.50	200
☐	Operational Efficiency	Supply Chain Optimization	9.00	7.50	1.50	267
☐	Operational Efficiency	Employee Training Enhancement	6.75	7.50	1.50	133
☐	Process Improvement	Automated Claims Processing	18.75	7.50	1.39	360

Figure 9 The Decision areas under the Project Selection Tab

Participants need to analyse these metrics and evaluate the trade-offs in selecting specific projects while considering the available budget. The decision-making process involves balancing risk, return, and strategic priorities to ensure an optimal project portfolio that maximizes value and aligns with business objectives.

2.6 Decision checklist

The Decision Checklist assists participants in reviewing and modifying their decisions before the deadline by consolidating them from each decision areas. Once decisions are finalized, participants can submit them using the Submit option if they opt to proceed with the Decision Checklist.

Decision Checklist		Round 2
		User
Project Portfolio Discount Rate		
Global Outreach Initiative		10%
Regional Domination Campaign		11%
Digital Transformation Drive		12%
Emerging Markets Expansion		0%
Efficiency Overhaul Program		0%
Supply Chain Optimization		0%
Employee Training Enhancement		0%
Automated Claims Processing		0%
Customer Onboarding Redesign		0%
Regulatory Compliance Enhancement		0%
Data Security Strengthening		0%
Innovative Insurance Products		0%
Partnership for New Market Entry		0%

Figure 10 The Decision Checklist

2.7 Report

The report section starts with the participants' critical thinking abilities in the food for thought section based on four metrics.

- Rigour: Consistency throughout the decision-making and recommendation.
- Structuring: Logical approach of gathering and analysing information
- Synthesis: Approach towards interpreting information and making informed decisions
- Business Judgement: Ability to evaluate facts, and consider risks and possible implications when making decisions.

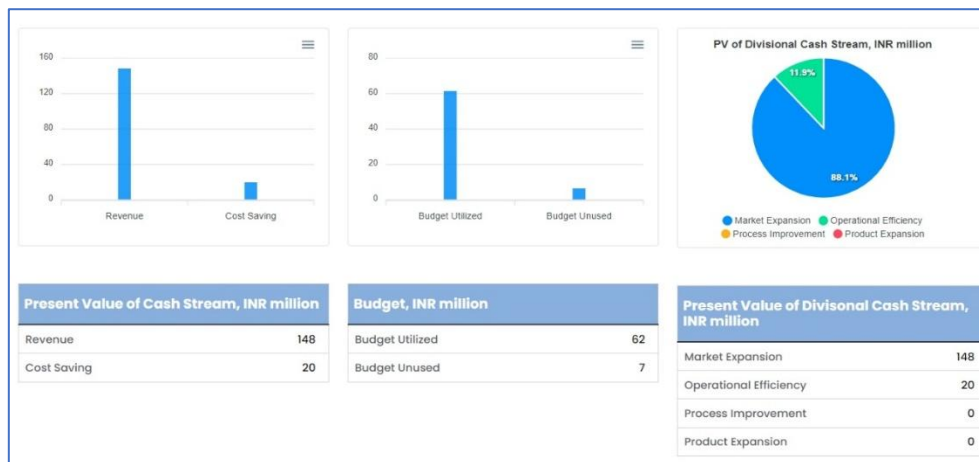


Figure 11 The Report section in the Game Arena

The Report section provides details on the Present Value of Divisional Cash Stream, revenue generated based on project selection, and cost savings. It also includes a table summarizing the various projects along with key financial metrics which are calculated based on the actual discount rate, offering a comprehensive financial overview of the selected projects.

Projects								
Area	Project	Investment, INR million	PV, INR million	NPV, INR Million	IRR %	Benefit Cost Ratio	Profitability Index	EAC
Market Expansion	Global Outreach Initiative	15	36	21	62%	2.4	1.4	4.3
Market Expansion	Regional Domination Campaign	11	48	37	128%	4.3	3.3	3.5
Market Expansion	Digital Transformation Drive	26	65	38	63%	2.5	1.5	7.1
Operational Efficiency	Supply Chain Optimization	9	20	11	55%	2.2	1.2	2.7

KPI	
Average discount rate across projects, %	9
Present value of cash Stream, INR million	168
Net present value of cash stream, INR million	107

Figure 12 The Report Section in the Game Arena

The KPI Section includes three key metrics that provide critical insights into the financial health and return potential of the selected projects, helping assess their overall value and impact on the portfolio.

1. Average Discount Rate Across Projects (%)

- This represents the weighted average discount rate applied to the selected projects.
- It reflects the risk-adjusted return expectation based on the mix of projects chosen.
- A lower discount rate generally indicates lower risk projects, while a higher rate suggests higher risk investments.

2. Present Value of Cash Stream

- This is the total value of all future cash inflows from the selected projects, discounted to today's value.
- It represents the expected earnings from the project portfolio adjusted for the time value of money.
- A higher present value means that the selected projects generate significant future cash flows.

3. Net Present Value (NPV) of Cash Stream

- NPV measures the profitability of the selected projects after accounting for the initial investment.
- A positive NPV indicates that the selected projects are expected to generate substantial net gains and add value to the company.

2.8 Synopsis

The synopsis section provides a summary of the decision areas and the report, highlighting how effectively you applied critical thinking in each aspect, outcomes in different areas, offering insights into the impact of your decision-making skills.

Synopsis		Round 1	XLS
Decisions		Present Value of Divisional Cash Stream, INR million	
Parameters	Input	Parameters	Output
Global Outreach Initiative	10.5%	Market Expansion	0
Regional Domination Campaign	10.5%	Operational Efficiency	20
Digital Transformation Drive	15.5%	Process Improvement	28
Emerging Markets Expansion	15.5%	Product Expansion	0
Efficiency Overhaul Program	12.5%		
Supply Chain Optimization	10.5%	Present Value of Cash Stream, INR million	
Employee Training Enhancement	12%	Parameters	Output
Automated Claims Processing	14.5%	Revenue	0
Customer Onboarding Redesign	11%	Cost Saving	48
Regulatory Compliance Enhancement	15.5%		
Data Security Strengthening	15.5%	Budget, INR million	
Innovative Insurance Products	0%	Parameters	Output
Partnership for New Market Entry	12.5%	Budget Utilized	30
Customizable Policy Options	12%	Budget Unused	38
Global Outreach Initiative	Not Selected		
Regional Domination Campaign	Not Selected	KPI	
Digital Transformation Drive	Not Selected	Parameters	Output
Emerging Markets Expansion	Not Selected	Average discount rate across projects, %	4%
Efficiency Overhaul Program	Not Selected	Present value of cash Stream, INR million	48
Supply Chain Optimization	Selected	Net present value of cash stream, INR million	18
Employee Training Enhancement	Not Selected		
Automated Claims Processing	Not Selected	Thinking Ability	
Customer Onboarding Redesign	Not Selected	Parameters	Output
Regulatory Compliance Enhancement	Not Selected	Project Portfolio	100%
Data Security Strengthening	Selected	Decide	75%
Innovative Insurance Products	Not Selected		
Partnership for New Market Entry	Not Selected		
Customizable Policy Options	Not Selected		

Round 2

To play the next iteration, follow these steps:

1. Go to the end of the synopsis page, to find the next round.
2. Click on the next round.
3. The participant will be redirected to the home page.
4. Click on the card to start the next round.

Note: The leaderboard for a round will only be updated only once after moving to the next round.